# Experiment 14

## Aim

To study and implement pro-code extension using Zoho Catalyst to extend the functionality of the Library Book Issue Management low-code application by developing custom backend logic, APIs, database operations, and advanced features that go beyond the built-in capabilities of Zoho Creator.

## Introduction

Zoho Catalyst is a pro-code development platform that enables developers to build, deploy, and manage applications using programming languages and cloud-based services. It supports backend functions, data storage, API development, authentication, and integrations.

Pro-code extension combines the rapid development capabilities of low-code platforms such as Zoho Creator with the flexibility of custom programming. This approach helps developers implement advanced business logic, complex calculations, and specialized integrations, for example a custom fine-calculation API for overdue library books.

## Procedure

1. Open the Zoho Catalyst website and sign in with your Zoho account.
2. Create a new Catalyst project (for example Library-Book-Issue).
3. Open CloudScale and explore the available components: Data Store, NoSQL, File Store, Authentication, API Gateway, and Cron.
4. Create a table in the Data Store (for example librarybookissue) and review its schema.
5. Install the Catalyst CLI (npm install -g zcatalyst-cli) and log in using catalyst login to develop and deploy functions from the terminal.
6. Connect the Catalyst function or API to the Zoho Creator application for advanced processing.

## Workflow Chart

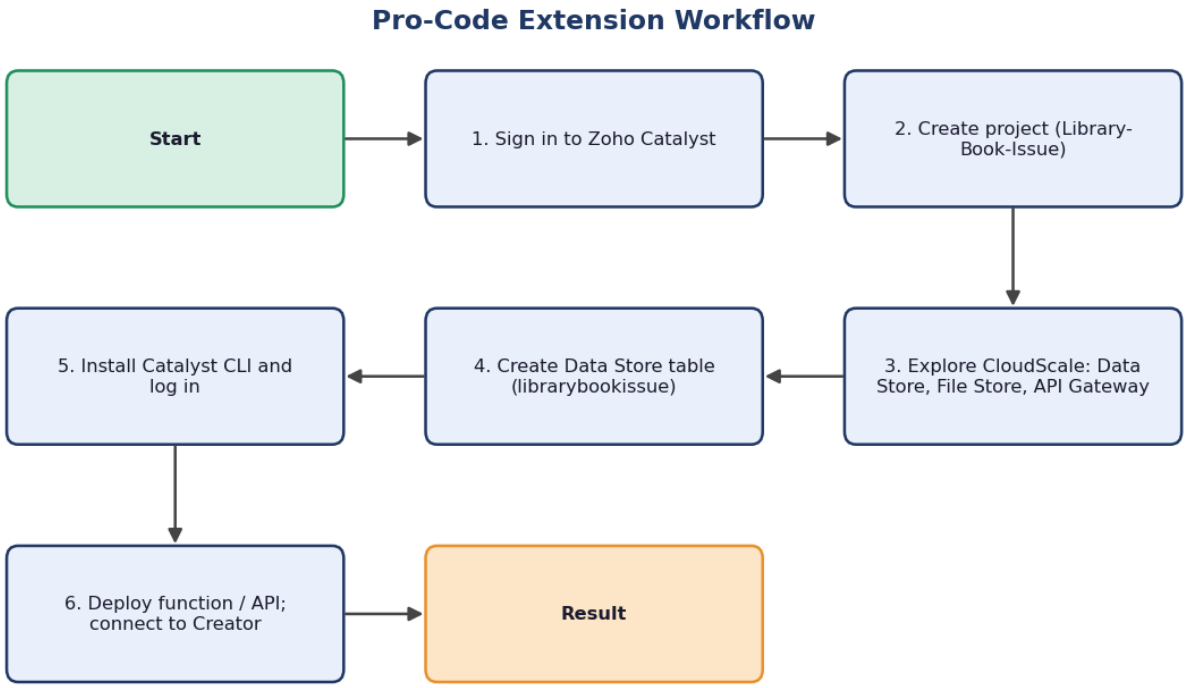



Figure 14-1: Pro-code extension workflow

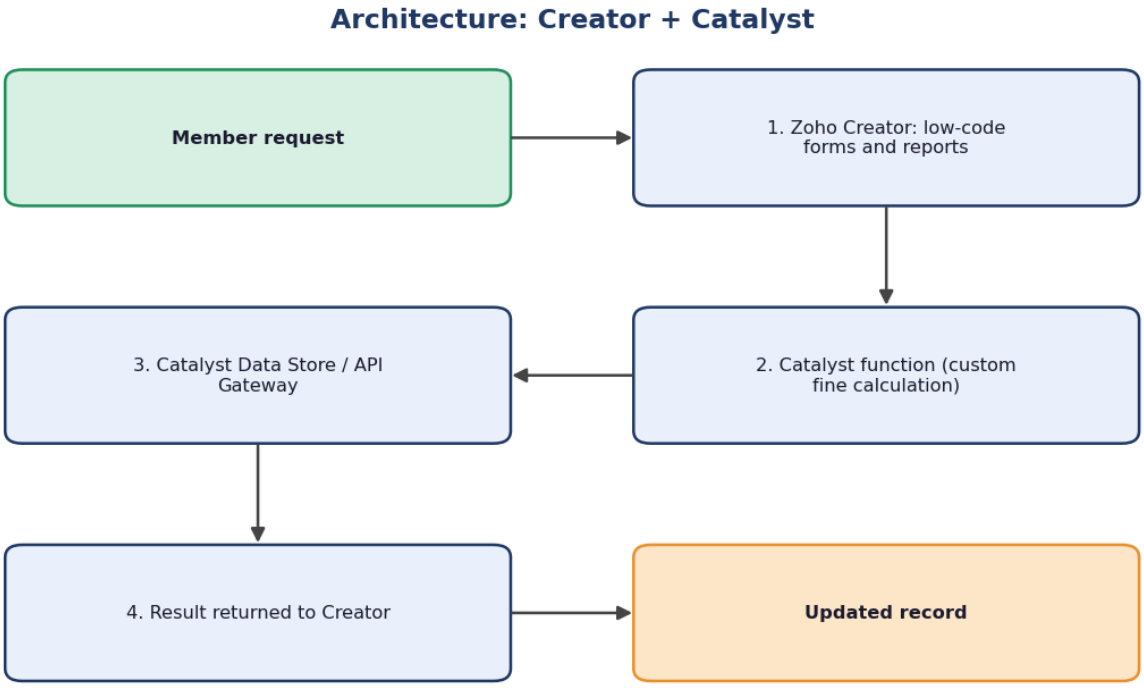



Figure 14-2: Architecture - Zoho Creator with Zoho Catalyst

## Conclusion

Pro-code extension using Zoho Catalyst combines the rapid development benefits of low-code platforms with the flexibility of traditional programming. It enables developers to create custom backend services, integrate external applications, and implement advanced functionality. This approach is useful when an application requires capabilities beyond the standard features available in Zoho Creator.